District-Level Model Performance: Figures 7a, 7b and 7c

What drives model quality across the 133 test-set districts?

1. The Unit of Analysis — Districts, Not Observations

Each point in figures 7a, 7b and 7c represents one unique livelihood zone (district), not an individual observation. The test set spans 7 temporal folds; a district contributes one row per fold it appears in, up to a maximum of 7. Across the 133 districts present in the test set, the median number of fold appearances is 6.

To compute a meaningful district-level PR-AUC, a minimum of 5 observations is required. This filter retains 74 of 133 districts (n = 74 in all three figures). The 59 excluded districts have too few appearances across folds to produce a reliable metric:

2 district(s) with 2 fold appearances
2 district(s) with 3 fold appearances
55 district(s) with 4 fold appearances

The retained districts are not a random sample — districts with more fold appearances tend to be those with more stable data pipelines and higher crisis activity, which itself shapes the patterns visible in the three scatter plots.

2. Fig 7a — News Volume vs ΔPR-AUC (does more coverage mean more benefit?)

x-axis: $\log_{10}$(mean GDELT articles per month per district), range: 1.97 – 4.47
y-axis: ΔPR-AUC = PR-AUC(AR+News) − PR-AUC(AR-Only), range: -0.800 – 0.417, median: -0.052

Pearson r:	**-0.018**
p-value:	**0.879 (not significant)**
OLS slope:	**-0.0064 (near zero)**
Article volume range:	**94 – 29,360 articles/month (log scale on figure)**

There is no relationship between how much GDELT coverage a district receives and how much it benefits from the news features. The regression line is essentially flat.

This is an important result. It rules out the simplest explanation for why news helps — that districts with more articles just give the model more data to work with. A district receiving 29,360 articles/month gains no more from news features than one receiving 94. What matters is not volume but whether the content carries a signal about crisis onset that is distinct from what the IPC history already captures. In many districts, even heavy news coverage adds nothing — the AR signal is already sufficient or the news pattern does not align with crisis timing.

3. Fig 7b — Volatility vs AR-Only PR-AUC (does regime switching help or hurt?)

x-axis: volatility = fraction of consecutive fold pairs where the regime changes, range: 0.17 – 1.00
y-axis: AR-Only PR-AUC, median: 0.621

Pearson r:	**0.132 (moderate positive)**
p-value:	**< 0.001 (highly significant)**
OLS slope:	**0.1636 (each 0.1 increase in volatility → +0.016 PR-AUC)**

Districts that switch between crisis and non-crisis states more frequently across folds show better AR-Only performance — not worse. This seems counterintuitive at first: a volatile district should be harder to predict.

The explanation is structural. Volatility implies the district has both crisis and non-crisis periods within the test window. This provides the model with a richer mix of positive and negative examples per district, giving PR-AUC more signal to measure against. A district that is always in crisis (zero volatility) produces a trivially perfect PR-AUC because every observation is positive — the model cannot get it wrong. A district that is always stable has no positives at all and is excluded from district-level PR-AUC computation. The moderate-volatility districts in the middle are the genuine prediction challenge, and the AR model performs well on them because the lag feature is most informative precisely when transitions exist to detect.

4. Fig 7c — Onset+Chronic Count vs AR-Only PR-AUC (does crisis frequency drive performance?)

x-axis: total onset + chronic crisis observations per district across all folds, range: 1 – 6
y-axis: AR-Only PR-AUC, median: 0.621

Pearson r:	**0.641 (strongest relationship of the three figures)**
p-value:	**< 0.001 (highly significant)**
OLS slope:	**0.1154 (each additional crisis fold → +0.115 PR-AUC)**

The strongest district-level predictor of AR model quality is simply how many times a district has been in crisis across the test folds. More crisis observations give the AR model more positive examples to learn from within that district's fold appearances.

There is also a direct statistical explanation: PR-AUC with only 1 positive observation is near-binary — it is either perfect (1.0, if that one case is scored highest) or very poor. Districts with 5 or 6 crisis observations across 7 folds produce more stable and informative PR-AUC estimates. This is not purely a statistical artefact, however — a district that appears in crisis frequently is also one where the AR lag feature (ipc_lag_1, ipc_persistence_2yr) has more sustained signal to anchor its predictions to.

The practical implication: the AR model is most reliable for districts with established crisis histories. In newly-deteriorating districts — those with few prior IPC crisis records — both the statistical estimate and the underlying prediction are weaker.

5. Reading the Three Figures Together

The three figures characterise what drives district-level model quality from three different angles — data volume, regime dynamics, and crisis frequency. The consistent story is:

News volume (7a) does not explain where news features help. Coverage quantity is not the bottleneck — signal quality and timing relative to IPC assessments are.

Volatility (7b) and crisis frequency (7c) both correlate strongly with AR-Only performance, and for related reasons: both are proxies for how much positive-class signal the AR model has access to within a district's history. Districts with richer, more varied crisis histories give the lag-based AR model more to work with.

Volatility does predict ΔPR-AUC, but negatively ($r = -0.566$, $p < 0.001$): districts where the AR model already performs well tend to gain less from news — there is less room for improvement. The flip side is that low-volatility districts, where the AR model struggles, are exactly where news has the most potential to add value.

Crisis frequency (onset+chronic count) does not predict ΔPR-AUC ($r = 0.072$, $p = 0.54$), nor does news volume ($r = 0.081$, $p = 0.50$). News benefit is not driven by how crisis-prone a district is or how much coverage it receives — it depends on whether GDELT coverage carries a pre-crisis signal that arrives before the IPC assessment does.